



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to N.N.C.FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

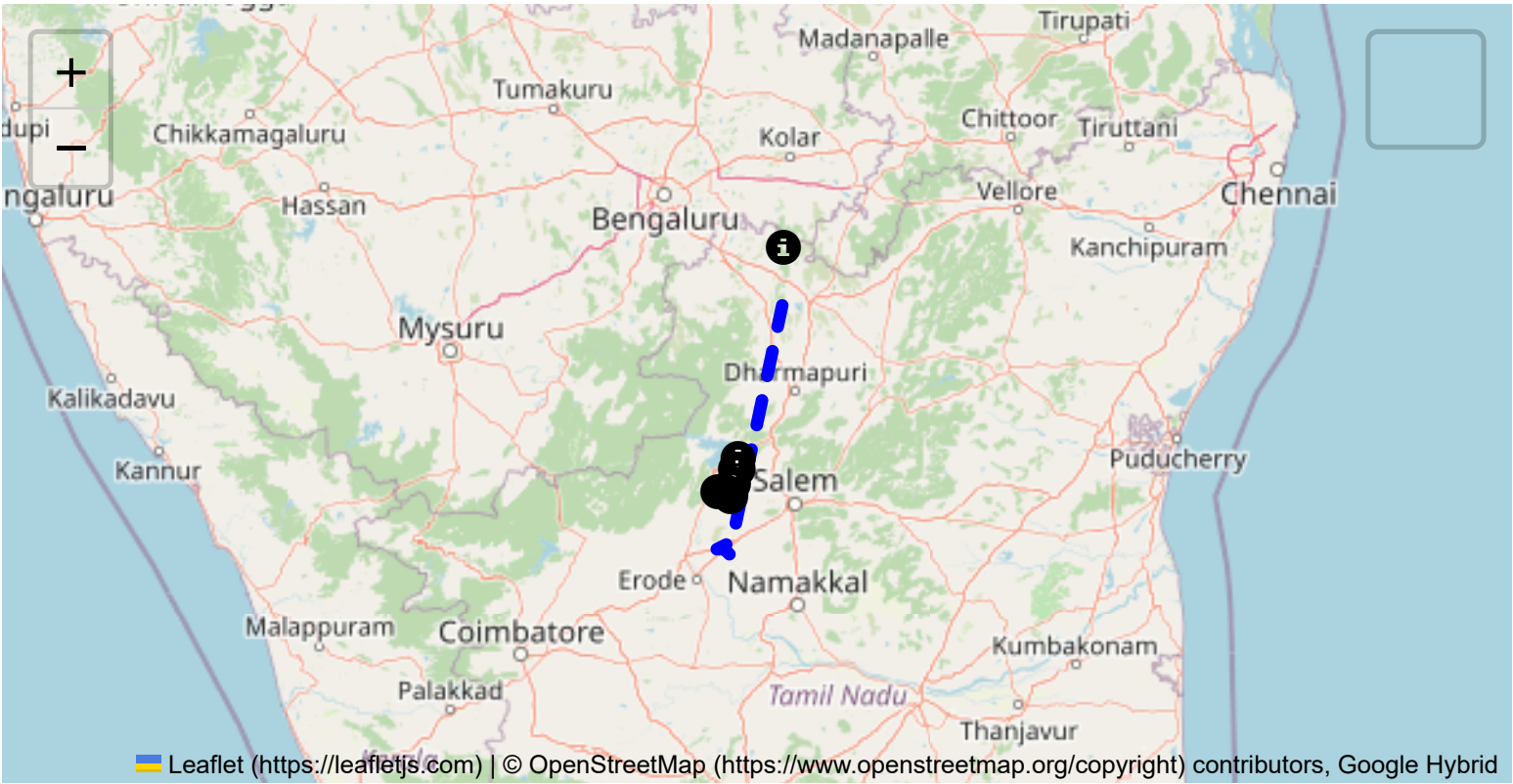
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

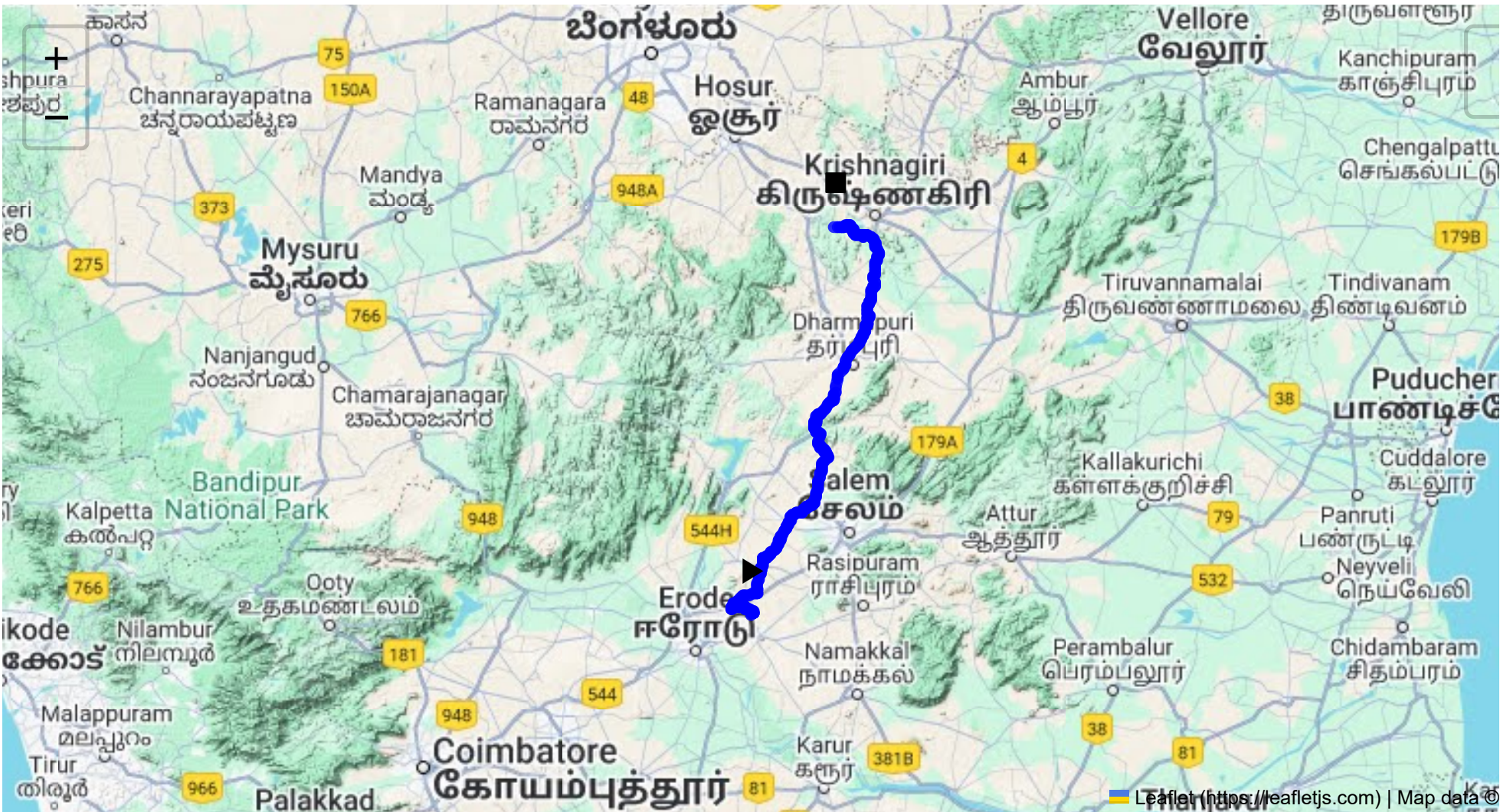
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 159.58 km
Estimated Duration: 3.2 hours
Adjusted Duration (Heavy Vehicle): 4.0 hours
Start: (11.4381, 77.8734)
End: (12.497201, 78.104324)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

This route covers approximately 159.58 kilometers from Sangagiri to Kodgur, spanning through several checkpoints including Sankari RS, Padaiveedu, Ayveli, Erumaipatti, and Koranampatti. The journey primarily follows SH 86, a vital link road in this region, facilitating regional transport.

2. Typical Weather Conditions and Potential Hazards

The typical weather in this region features a tropical climate with moderate to high temperatures year-round. The area experiences monsoon rains from June to September which can lead to flooding and reduced visibility on the roads, thus increasing the risk of skidding and accidents. Post-monsoon, fog can be present in the early mornings, affecting visibility.

3. Analysis of Traffic Patterns

Traffic congestion is moderate in Sangagiri and Sankari during peak hours (8-10 AM and 5-8 PM). The Bhavani Main Road section often encounters traffic due to agricultural and local transportation activities. SH 86 experiences less congestion, but minor holdups can occur near major town junctions.

4. Assessment of Road Quality and Infrastructure

The quality of the roads varies; SH 86 is generally well-maintained but may have potholes or uneven surfaces post-monsoon. Infrastructure like signage and lighting is adequate near towns but diminishes in rural stretches. Ongoing construction can occur without warning, necessitating caution.

5. Suggestions for Alternative Routes for Emergencies

In case of road closures, NH 544 (near Sangagiri) can be used as an alternate route looping north, reconnecting to SH 86 post-Kondenadu. Always consult with local authorities for updates.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to the Motor Vehicles Act and the Central Motor Vehicles Rules in India. Drivers must carry appropriate documentation and utilize specified routes. Night travel with hazardous materials requires prior permission in certain regions.

7. Overview of Historical Incidents

There have been few recorded incidents specifically involving heavy vehicles or hazardous materials on this route; however, general accidents are often reported near Sankari and Ayveli due to sharp turns and speeding.

8. Environmental Considerations and Sensitive Areas

This route crosses through agricultural lands, requiring attention to prevent spills or emissions. Nearby water bodies warrant careful navigation to avoid contamination.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable in towns but can be patchy in rural areas between major townships, posing potential communication challenges.

10. Estimated Emergency Response Times

Emergency services in towns like Sangagiri and Sankari have faster response times, averaging 15-30 minutes, whereas rural sections could face delays up to 60 minutes due to distance and road conditions.

11. Overall Summary of Risk Assessment

Overall, while the route is moderately safe for transportation of hazardous materials, it poses risks primarily from weather-related issues, road quality variances, and inconsistent communication coverage. Adherence to regulations and maintaining safe driving practices, especially in known congestion and weather-sensitive areas, will mitigate risks. Awareness of alternate routes and local traffic patterns can aid in efficient navigation and emergency preparedness.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	Medium	11.45654, 77.81706	30 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	High	11.64243, 77.95198	15 KM/Hr
16	Turn	Medium	11.64721, 77.95118	30 KM/Hr
17	Turn	Medium	11.67870, 77.96846	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71416, 77.99604	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	Medium	11.71715, 77.99969	30 KM/Hr
21	Turn	Medium	11.71921, 78.01133	30 KM/Hr
22	Turn	Medium	11.72737, 78.02248	30 KM/Hr
23	Turn	High	11.73775, 78.03465	15 KM/Hr
24	Turn	High	11.73651, 78.03878	15 KM/Hr
25	Turn	High	11.74189, 78.04660	15 KM/Hr
26	Turn	Medium	11.78039, 78.05792	30 KM/Hr
27	Turn	Medium	11.86753, 78.08331	30 KM/Hr
28	Turn	Medium	11.86981, 78.08387	30 KM/Hr
29	Turn	Medium	11.88041, 78.08143	30 KM/Hr
30	Turn	Medium	11.91302, 78.05856	30 KM/Hr
31	Turn	Medium	11.92526, 78.06488	30 KM/Hr
32	Turn	Medium	11.93402, 78.06354	30 KM/Hr
33	Turn	Medium	11.94600, 78.04740	30 KM/Hr
34	Turn	Medium	11.95625, 78.04748	30 KM/Hr
35	Turn	High	11.95880, 78.04989	15 KM/Hr
36	Turn	High	11.96040, 78.04947	15 KM/Hr
37	Turn	High	11.96320, 78.05139	15 KM/Hr
38	Turn	Medium	11.96463, 78.05077	30 KM/Hr
39	Turn	High	11.96597, 78.04920	15 KM/Hr
40	Turn	Medium	11.98504, 78.06287	30 KM/Hr
41	Turn	High	12.24611, 78.19519	15 KM/Hr
42	Turn	Medium	12.25200, 78.20034	30 KM/Hr
43	Turn	High	12.40026, 78.21775	15 KM/Hr
44	Turn	Medium	12.40734, 78.22344	30 KM/Hr
45	Turn	High	12.46292, 78.21416	15 KM/Hr
46	Turn	High	12.47472, 78.19192	15 KM/Hr
47	Turn	Medium	12.47225, 78.19035	30 KM/Hr
48	Turn	High	12.47108, 78.18638	15 KM/Hr
49	Turn	High	12.47277, 78.18497	15 KM/Hr
50	Turn	Medium	12.47214, 78.18393	30 KM/Hr
51	Turn	High	12.47280, 78.17768	15 KM/Hr
52	Turn	High	12.47511, 78.17676	15 KM/Hr
53	Turn	Medium	12.47578, 78.17449	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
54	Blind Spot	Blind Spot	12.47678, 78.15977	10 KM/Hr
55	Turn	High	12.47849, 78.16027	15 KM/Hr
56	Turn	Medium	12.48756, 78.15444	30 KM/Hr
57	Blind Spot	Blind Spot	12.48912, 78.15137	10 KM/Hr
58	Blind Spot	Blind Spot	12.49064, 78.15282	10 KM/Hr
59	Turn	Medium	12.49234, 78.15098	30 KM/Hr
60	Turn	High	12.49576, 78.15116	15 KM/Hr
61	Turn	Medium	12.49986, 78.14721	30 KM/Hr
62	Turn	High	12.49436, 78.12425	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
14	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
29	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
30	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45654, 77.81706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59214, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67870, 77.96846



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71416, 77.99604



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72737, 78.02248



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73775, 78.03465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03878



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74189, 78.04660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78039, 78.05792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92526, 78.06488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93402, 78.06354



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94600, 78.04740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95625, 78.04748



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95880, 78.04989



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96040, 78.04947



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96463, 78.05077



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96597, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98504, 78.06287



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.24611, 78.19519



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25200, 78.20034



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.40026, 78.21775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40734, 78.22344



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.46292, 78.21416



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.49436, 78.12425

